

Pau Petit Rosàs

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PhD researcher in theoretical particle physics with expertise in scattering amplitudes, Feynman integrals, differential equations and Monte Carlo generators. Proven experience addressing unresolved quantitative problems by building high-performance numerical tools in Python, C++, and Fortran, with research presented at leading scientific conferences and workshops worldwide.

EDUCATION

PhD in Theoretical Physics , University of Liverpool	Oct 2023 – Present
MPhys in Theoretical Physics , University of Manchester, First Class	Sep 2019 – Jun 2023
Spanish Baccalaureate , Ginebró High School, 97%, Top of class	Sep 2017 – Jun 2019

PUBLICATIONS – [SCIENTIFIC PROFILE](#)

Publications: 7 **Citations:** 434 **First-author:** 4

1. Dave, Thomas et al. (Apr. 2026). “Tensor decomposition of $e^+e^- \rightarrow \pi^+\pi^-\gamma$ to higher orders in the dimensional regulator”. In: arXiv: [2604.16251 \[hep-ph\]](#).
2. Petit Rosàs, Pau, Olga Shekhovtsova, and William J. Torres Bobadilla (2026). “Radiative return meets GVMD”. In: *Phys. Lett. B* 881, p. 140864. DOI: [10.1016/j.physletb.2026.140864](#). arXiv: [2603.13171 \[hep-ph\]](#).
3. Petit Rosàs, Pau (Mar. 2026). “Evaluation of Feynman integrals via numerical integration of differential equations”. In: *17th International Symposium on Radiative Corrections: Applications of Quantum Field Theory to Phenomenology*. arXiv: [2603.05336 \[hep-ph\]](#).
4. Petit Rosàs, Pau et al. (2026). “Phokhara at the frontier of NNLO”. In: *Proceedings of The European Physical Society Conference on High Energy Physics — PoS(EPS-HEP2025)*. Vol. 485, p. 294. DOI: [10.22323/1.485.0294](#).
5. Petit Rosàs, Pau and William J. Torres Bobadilla (2025). “Fast evaluation of Feynman integrals for Monte Carlo generators”. In: *JHEP* 09, p. 210. DOI: [10.1007/JHEP09\(2025\)210](#). arXiv: [2507.12548 \[hep-ph\]](#).
6. Aliberti, R. et al. (2025). “The anomalous magnetic moment of the muon in the Standard Model: an update”. In: *Physics Reports* 1143, pp. 1–158. ISSN: 0370-1573. DOI: [10.1016/j.physrep.2025.08.002](#). URL: <http://dx.doi.org/10.1016/j.physrep.2025.08.002>.
7. Aliberti, Riccardo et al. (2024). “Radiative corrections and Monte Carlo tools for low-energy hadronic cross sections in e^+e^- collisions”. In: *SciPost Physics Community Reports*. URL: <http://dx.doi.org/10.21468/SciPostPhysCommRep.9>.

AWARDS

- **Leverhulme Trust** – Grant funding the PhD stipend and travel expenses.
- **Tessella Prize** – Best use of software in a master’s thesis, University of Manchester.
- **Gold Medal** – Catalan National League of Debate.

SKILLS

Python	Pandas, NumPy, SciPy
C++	MPI & OpenMPI, BOOST, ODEINT
Fortran	High precision, optimization
Linux	Bash, Git, Clusters

LANGUAGES

English	Fluent
Spanish	Native
French	B1
Catalan	Native